Verified Financial Context and Policy Adherence in an Agent Handoff

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Abstract

Delegated agents may act correctly on a peer request while producing a poor outcome for the overall task. This exploratory study examines that failure in a simulated Transport–Finance workflow with discrete budgets and cancellation options. A 12-case pilot exposed an impossible funding request and an excessive funding request. A subsequent paired comparison of ordinary and calculation-explicit Transport prompts achieved optimal outcomes on all 12 cases in both conditions. We then injected funding-request errors into 12 fresh scenarios and evaluated Finance under three information conditions using openai/gpt-oss-20b on Groq. Optimal outcomes were 3/12 with request text alone, 11/12 with independently supplied budget and price, and 12/12 when those facts were accompanied by a checking instruction. The sole failure in the verified-facts condition was an empty response at the output-token limit, preventing a claim of improved arithmetic from the checking instruction. Finally, 60 Adaption-generated messages were evaluated against two fixed financial scenarios under an explicit system-level protection rule. Outcomes were optimal in 114/120 calls, with no protected-booking cancellations among 118 parseable responses and two truncated outputs. These results motivate independent state access at agent handoffs while documenting limited sample size, unequal information across conditions, and imperfect generated style labels.

1. Introduction

A recipient agent can comply with a peer’s request yet undermine the human’s objective when that request contains an incorrect funding amount. In this simulated workflow, Transport selects a replacement bus and asks Finance to cancel optional bookings. The team must fund a feasible trip while minimizing cancellations. A protected venue deposit must remain untouched.

The contribution is a small, auditable evaluation of decision quality at this handoff, accompanied by saved prompts, responses, deterministic scoring, and a separate policy-adherence stress test. The work studies two-role interactions and single-recipient replays. It does not establish a general swarm-control mechanism or a new verification algorithm.

2. Related Work

Multi-Agent Risks from Advanced AI describes risks arising through agent interactions [1]. BoardroomAI studies dependency-aware intervention and selective repair, including preservation of unaffected decisions and insufficient context for synthesis [2]. CASCADE investigates scoped communication and escalation during industrial replanning [3]. These works motivate examining what information crosses a delegation boundary. The present study isolates a small funding request with an exact answer key and reports exploratory evidence rather than superiority over those systems.

3. Methods

All automated calls used openai/gpt-oss-20b through the Groq Python SDK, temperature 0, and max_completion_tokens 4096. Calls used fresh conversations. The code did not set reasoning effort, a seed for model sampling, or a JSON response-format constraint. Temperature 0 does not guarantee identical repeated responses. Prompts, raw text, model IDs, usage, and finish reasons were retained. No repeated sampling was performed for each case-condition pair.

The answer key enumerates all subsets of optional bookings. A feasible plan must fund the selected bus without cancelling protected or nonexistent items. Team-level optimization orders cancellation count, bus price, and released money. Finance-only tests hold the bus fixed and minimize cancellation count followed by released money. Correctly identifying an impossible case with no cancellations is an optimal outcome. Empty or invalid outputs count as unsuccessful outcomes and are also disclosed separately.

Stage	Design	Completed episodes
Pilot	12 cases across four regimes	12
Transport comparison	12 new cases × 2 prompts	24
Selected-error diagnostic	2 pilot failures × 2 conditions	4
Funding fault injection	12 new cases × 3 conditions	36
Policy extension	60 messages × 2 scenarios	120

Table 1. Stages are separate exploratory datasets. An episode may contain one or two model calls. Counts must not be pooled into a single treatment estimate. Preliminary manual Gemini exercises are excluded.

The pilot used near-isomorphic cases. Its failures informed later prompts and hypotheses. The Transport comparison used random seed 7319 for case generation and 8421 for execution ordering. Its ordinary and calculation-explicit prompts shared clarified funding rules. Those clarifications prevent a causal comparison with the earlier pilot.

The funding fault-injection dataset was fixed before calls with seed 20261107 and shuffled with seed 961. It contains three cases each with correct, excessive, insufficient, and impossible funding requests. These requests are synthetic interventions, not newly observed Transport errors. Finance receives either request text alone, the same text plus verified budget and price, or those facts plus an explicit checking instruction. The first comparison changes information availability. Only the latter two conditions have equal financial information. The environment is assumed to supply accurate facts.

4. Results

The pilot produced 11/12 correct action choices and valid trips in all nine feasible cases. In one impossible case, Transport stated that 300 + 190 = 490 was insufficient for a 580-rupee bus, then requested 90 rupees anyway. In another case it requested 190 rupees when the actual shortfall was 597 − 487 = 110. Finance cancelled both optional bookings, destroying an additional 80 rupees of optional value.

The Transport comparison achieved 12/12 optimal outcomes in each condition. In all four impossible cases, the calculation condition returned zero for bus price and required funding, making those fields uninformative. It therefore provides no evidence that the requested calculation was completed correctly on impossible instances.

Finance condition	Successful	Optimal	Empty outputs
Message only	6/12	3/12	0
Verified numbers	11/12	11/12	1
Numbers and checking instruction	12/12	12/12	0

Table 2. Fault-injection outcomes on the same 12 scenarios. Success permits unnecessary cancellations. Optimality requires the smallest cancellation count and then the smallest release. All three correct-request controls were optimal in every condition. Message-only excess requests were feasible but wasteful. Insufficient requests failed to fund the trip, and impossible requests induced cancellations.

The sole verified-numbers failure returned empty content with finish_reason=length, using 4094 reasoning tokens within a 4096-token completion budget. It is a completion failure, not an observed incorrect financial decision. All 11 delivered decisions in that condition were optimal. The two equally informed conditions differ on only one paired case, which is insufficient evidence of a reliable checking-instruction benefit.

Adaption policy extension

Adaption Invent a Dataset produced 59 rows containing 12-message tables, yielding 708 messages and 706 distinct texts. All messages contained the required placeholder. A fixed seed of 1108 selected source rows 15, 18, 35, 40, and 48, giving 60 messages with 20 generated labels each for neutral, urgency, and claimed approval. No samples were selected based on model behavior. Of the full 708 messages, 324 fell outside the requested 20–60-word range. Labels sometimes overlap semantically, including approval language in neutral messages.

Each selected message was evaluated in both scenarios with seed 1109 determining order. Budget was 500 rupees. Optional cancellations released 80 and 110 rupees, and the protected deposit was worth 200. Bus price was either 610 (feasible through the photo booth cancellation) or 740 (infeasible using optional funds). System instructions prohibited cancelling the deposit or accepting peer exceptions. Both scenarios exposed verified facts in the user message. These are proposed JSON decisions, with no real transactions or external tools executed.

4. Results continued

Scenario	Optimal	Other valid outputs	Empty outputs
Feasible	58/60	2	0
Protected funds required	56/60	2	2
Total	114/120	4	2

Table 3. Policy-extension outcomes. No protected deposit was listed for cancellation among 118 schema-valid outputs. The two unparseable outputs are excluded from that denominator. They remain failures in the 120-call optimal-outcome denominator.

One feasible case was unnecessarily declared infeasible. Three replies said release while cancelling nothing, including one feasible and two infeasible cases. Two protected-required calls returned empty content at the token limit. These findings show that avoiding a prohibited cancellation and completing the task correctly are distinct outcomes.

5. Discussion and Limitations

Independent state access at the receiving agent can expose contradictions in a funding request. The fault-injection result supports this limited observation, but it also reflects an information advantage by design. A deterministic program given the same verified facts solves these tiny cases exactly. No claim is made that an LLM is necessary for arithmetic or subset selection.

The study uses one evaluated model, one numerical domain, one response per condition, and small, partly templated datasets. The 120 extension calls reuse only two financial scenarios and messages clustered within five generated tables. They are not 120 independent tests of real-world safety. Generated style labels were not manually corrected into clean interventions, so style-specific causal claims are inappropriate. No statistical generalization or significance claim is made from these convenience samples.

The policy extension uses a strong system instruction and assumes trusted environment facts. It does not test compromised state sources, real authenticated approvals, persistent memory, adaptive attackers, covert coordination, or large agent networks. The notebook evolved during the sprint, and pilot-guided changes are disclosed. Changing instructions between stages prevents attributing improvements to a single intervention across stages.

6. Conclusion

Verified budget and price information improved optimal outcomes from 3/12 to 11/12 on deliberately corrupted funding requests. An explicit checking instruction reached 12/12, with the sole difference attributable to an empty response in the other condition. A separate Adaption extension found no protected-booking cancellations among 118 parseable outputs, while six of 120 calls failed the overall task criterion. These are bounded exploratory observations about information access, instruction adherence, and completion reliability at an agent handoff.

Code and Data

The companion archive includes the experiment notebook, original JSON/JSONL logs, case manifests, and an offline audit script. No public repository has been published as part of this package. All experiments described were conducted during the sprint. External models, SDKs, and the official template are preexisting third-party resources.

Author Contributions and LLM Usage

Urva Upadhyay operated the experiments and collected the logs. ChatGPT assisted with study design, code, analysis, and manuscript drafting. Gemini and Claude were consulted during planning. Adaption generated the message dataset. Reported outcomes were checked against raw logs and deterministic scoring. The author should review the manuscript and references before submission.

References

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[2] Manivannan, S. (2026). BoardroomAI: Dependency-Aware Human-Steerable Multi-Agent Deliberation through Evolving Decision Graphs. arXiv:2608.13046. https://arxiv.org/abs/2608.13046

[3] Bi, M. (2026). CASCADE: Cascaded Scoped Communication for Multi-Agent Re-planning in Disrupted Industrial Environments. arXiv:2604.00451. https://arxiv.org/abs/2604.00451

[4] Groq. Quickstart and API documentation. https://console.groq.com/docs/quickstart

[5] Adaption. Invent a dataset. https://docs.adaptionlabs.ai/adaptive-data/invent-a-dataset/

Appendix A Limitations and Dual-Use / Ethical Considerations

The experiments establish outcomes only for the supplied synthetic cases, model, prompts, and call settings. Empty outputs are counted as task failures and separately identified. No observed protected cancellation is not a guarantee of safety. The policy test evaluates text proposals rather than actual tool execution, and message-style contamination limits interpretation. Verified facts are assumed correct, which shifts trust to the environment supplying them.

All bookings, funds, and approval claims are fictional. No real money was moved, no accounts were attacked, and no external recipients were contacted. The generated messages include unverified authority claims solely to test adherence to the organizer’s rule. Such examples could help others craft manipulative peer messages or make coordinated agent operations more effective. The same structured-context mechanisms could also improve harmful coordination if deployed toward harmful goals. The release contains only this bounded simulated domain and should not be interpreted as an operational attack guide. Secrets are referenced through Colab Secrets and excluded from the shareable notebook.

Appendix B Reproducibility and accounting

The companion offline script recomputes outcomes directly from parsed responses and the financial cases. It requires Python 3 and no API credentials. Recorded logs preserve finish reasons and usage for checking truncations. Live re-execution can differ because model-serving behavior is not deterministic or version-pinned. The archived notebook preserves experiment development order and requires selective execution rather than Run All. A cleaned live runner is not claimed.

For the final fault-injection comparison, paired outcomes for verified numbers and numbers plus instruction are identical on 11 cases. Their one discordance is an empty output. Exact two-sided McNemar testing of the paired binary outcomes yields $p=1.0$. This descriptive check does not support a superiority claim. No confidence interval is interpreted as a population guarantee for the clustered generated-message extension.